KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION

CHAPTER 1

YEAR 10 MATHEMATICS

# Financial mathematics

7 lessons · two pages each · Easy, then Medium


1.1 Percentages, fractions and decimals

1.2 Percentage increase and decrease

1.3 Profit, mark-up and discount

1.4 Wages and overtime

1.5 Commission, gross and net income

1.6 Simple interest and loans

1.7 Compound interest


NAME _____ CLASS _____ TEACHER _____

Artwork: gold spirals deep in Seahorse Valley, coloured in gold and green, part of the Mandelbrot set.

## How to use this booklet

1. Each lesson is **two pages**. Read the **learning intention** and tick each **success criterion** when you can do it.

2. Page 1 is **Easy** and page 2 is **Medium**. Each page starts with an example, then questions for you.

**WE DO** **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

**YOU DO** **Blue boxes.** Your turn. Work down the box one round at a time. Answers go on the line; show working in the boxes.

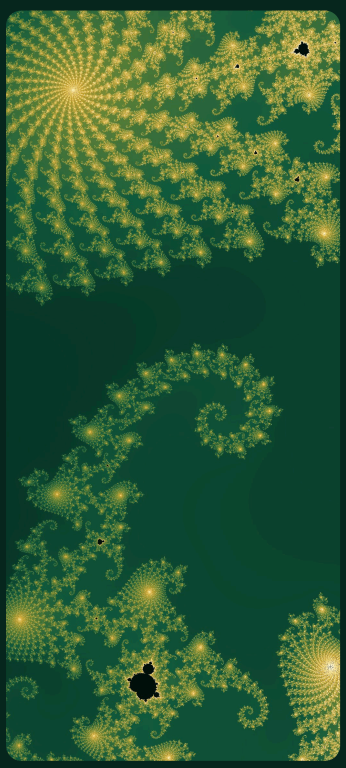
3. Stuck? Read the **Remember** line at the top of the blue box, then look back at the example you copied.

## Contents and progress tracker


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## In this chapter you will

- change between percentages, fractions and decimals
- find a percentage of an amount, and increase or decrease an amount by a percentage
- work out profit, mark-ups and discounts
- calculate wages, overtime, commission and net income
- calculate simple interest and compound interest



### About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is gold spirals deep in Seahorse Valley, coloured in gold and green. Maths can make beautiful things.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON 1.1

# Percentages, fractions and decimals

Date _____

**Learning intention**
We are learning to change between percentages, fractions and decimals, and find a percentage of an amount.

**Success criteria**
- ☐ I can change a percentage to a decimal or a fraction, and back again
- ☐ I can find a percentage of an amount of money
- ☐ I can write a score as a percentage

## Level 1 · Getting started

Easy · 32 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** **Change each one. Percent means "out of 100".**

**a** Write 45% as a decimal.

**b** Write 0.6 as a percentage.

**c** Write 20% as a fraction in simplest form.

**d** Write $\frac{3}{4}$ as a percentage.

**YOU DO** On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER % → decimal: ÷ 100 · Decimal → %: × 100 · % → fraction: put it over 100, then simplify · Fraction → %: × 100.

**ROUND 1** Questions 1–8: write each percentage as a decimal.

1. 50%
2. 25%
3. 70%
4. 9%
5. 36%
6. 100%
7. 4%
8. 85%

**ROUND 2** Questions 9–16: write each decimal as a percentage.

9. 0.3
10. 0.75
11. 0.05
12. 0.62
13. 0.9
14. 0.01
15. 1
16. 0.48

**ROUND 3** Questions 17–24: write each percentage as a fraction in simplest form.

17. 50%
18. 25%
19. 10%
20. 75%
21. 40%
22. 30%
23. 1%
24. 60%

**ROUND 4** Questions 25–32: write each fraction as a percentage.

25. $\frac{1}{2}$
26. $\frac{1}{4}$
27. $\frac{1}{10}$
28. $\frac{3}{4}$
29. $\frac{1}{5}$
30. $\frac{7}{10}$
31. $\frac{2}{5}$
32. $\frac{9}{100}$

## Level 2 · Building up

Medium · 28 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Use a calculator. Show the sum you type in.**

**a** Find 10% of $80.

**b** Find 25% of $36.

**c** Find 15% of $240.

**d** Write 21 out of 60 as a percentage.

**YOU DO** On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

28 QUESTIONS

**REMEMBER** % of an amount: change the % to a decimal, then × the amount (15% of $240 = 0.15 × 240) · Score as a %: score ÷ total × 100.

### ROUND 1 Questions 1–8: find each amount.

1. 10% of $70
2. 50% of $46
3. 25% of $200
4. 20% of $35
5. 5% of $80
6. 1% of $900
7. 75% of $40
8. 30% of $150

### ROUND 2 Questions 9–16: use a calculator to find each amount.

9. 15% of $60
10. 12% of $350
11. 35% of $80
12. 8% of $125
13. 45% of $120
14. 6% of $250
15. 28% of $89
16. 2.5% of $400

### ROUND 3 Questions 17–24: write each test score as a percentage.

17. 12 out of 20
18. 18 out of 25
19. 21 out of 60
20. 45 out of 50
21. 7 out of 10
22. 30 out of 40
23. 33 out of 60
24. 34 out of 40

### ROUND 4 Questions 25–28: solve each problem. Show your working.

25. Mia scored 85% on a test marked out of 40. How many marks did Mia get?
26. Sam saves 10% of a $450 pay. How much does Sam save?
27. A school has 800 students. 35% of them catch the bus. How many students catch the bus?
28. Jay earned $600 and spent $150 on rent. What percentage of the money went on rent?

LESSON 1.2

# Percentage increase and decrease

Date _____

**Learning intention**
We are learning to increase and decrease an amount by a percentage, and work with GST.

**Success criteria**
- ☐ I can increase or decrease an amount by a percentage
- ☐ I can use a multiplier such as × 1.08 or × 0.92
- ☐ I can add GST to a price, and find the GST in a price

## Level 1 · Getting started

Easy · 32 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the percentage first. Then add it on (increase) or take it away (decrease).

**a** Increase $40 by 10%.

**b** Decrease $60 by 10%.

**c** Increase $80 by 25%.

**d** Decrease $200 by 5%.

**YOU DO** On your own: Easy work down the box, one round at a time. Write each answer on its line.

32 QUESTIONS

**REMEMBER** Find the % of the amount (10% of $40 = $4) · Increase: add it on ($44) · Decrease: take it away ($36).

**ROUND 1** Questions 1–8: increase each amount by 10%.

1. $20
2. $70
3. $30
4. $120
5. $90
6. $250
7. $500
8. $15

**ROUND 2** Questions 9–16: decrease each amount by 10%.

9. $50
10. $80
11. $20
12. $300
13. $70
14. $140
15. $400
16. $35

**ROUND 3** Questions 17–24: increase or decrease each amount.

17. Increase $40 by 50%
18. Decrease $80 by 25%
19. Increase $60 by 20%
20. Decrease $300 by 5%
21. Increase $100 by 25%
22. Decrease $30 by 50%
23. Increase $400 by 5%
24. Decrease $50 by 20%

**ROUND 4** Questions 25–32: a price goes up or down. Find the new price.

25. $60, up 10%
26. $40, down 25%
27. $120, up 50%
28. $90, down 20%
29. $20, up 5%
30. $200, up 25%
31. $16, down 50%
32. $500, down 5%

## Level 2 · Building up

Medium · 28 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Use a multiplier and a calculator. Show the sum you type in.**

**a** Increase $20 by 8%.

**b** Decrease $20 by 8%.

**c** Add 10% GST to $70.

**d** Find the GST in $110 (GST inc.).

**YOU DO** On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

28 QUESTIONS

**REMEMBER** Increase by 8%: × 1.08 · Decrease by 8%: × 0.92 · Add GST: × 1.1 · GST in a price (GST inc.): ÷ 11.

**ROUND 1** Questions 1–8: write the multiplier for each change.

1. Increase by 15%
2. Decrease by 30%
3. Increase by 20%
4. Decrease by 5%
5. Increase by 50%
6. Decrease by 12%
7. Increase by 3%
8. Decrease by 25%

**ROUND 2** Questions 9–16: use a multiplier to find each new amount.

9. $80, up 15%
10. $250, down 12%
11. $968, up 12%
12. $4 900, down 7%
13. $45, up 6%
14. $360, down 35%
15. $1 200, up 3%
16. $89, down 20%

**ROUND 3** Questions 17–20: add 10% GST to each price.

17. $40
18. $90
19. $250
20. $18

**ROUND 4** Questions 21–24: each price includes GST. Find the GST in it.

21. $22
22. $44
23. $99
24. $330

**ROUND 5** Questions 25–28: solve each problem. Show your working.

25. A price tag says "$55 (GST inc.)". How much GST is included in the price?
26. Ali earns $25 an hour. Ali gets a 4% pay rise. What is Ali's new pay per hour?
27. A car is worth $18 000. Its value goes down by 15% in one year. What is the car worth now?
28. A school has 900 students. The number goes up by 6%. How many students are there now?

LESSON 1.3

# Profit, mark-up and discount

Date _____

**Learning intention**
We are learning to work out profit, mark-ups and discounts, in dollars and as percentages.

**Success criteria**
- ☐ I can find the profit or loss when something is sold
- ☐ I can find the new price after a mark-up or a discount
- ☐ I can write a profit as a percentage of the cost price

## Level 1 · Getting started

Easy · 32 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** **Cost price = what the shop pays. Selling price = what it sells for.**

**a** Cost $12, sold $20. Find the profit.

**b** Cost $50, sold $35. Is it a profit or a loss? How much?

**c** Cost $40, mark-up $15. Find the selling price.

**d** Price $90, discount $25. Find the sale price.

**YOU DO** On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

32 QUESTIONS

**REMEMBER** Profit = selling price − cost price (if it sells for less than it cost, it is a loss) · Selling price = cost price + mark-up · Sale price = price − discount.

**ROUND 1** Questions 1–8: find the profit.

1. Cost $20, sold $35
2. Cost $30, sold $45
3. Cost $50, sold $80
4. Cost $25, sold $40
5. Cost $100, sold $150
6. Cost $8, sold $15
7. Cost $60, sold $95
8. Cost $200, sold $260

**ROUND 2** Questions 9–16: find the profit or the loss. Write "profit" or "loss".

9. Cost $40, sold $55
10. Cost $30, sold $20
11. Cost $90, sold $70
12. Cost $15, sold $25
13. Cost $120, sold $100
14. Cost $65, sold $80
15. Cost $200, sold $150
16. Cost $45, sold $60

**ROUND 3** Questions 17–24: add the mark-up to the cost price to find the selling price.

17. $20, mark-up $5
18. $40, mark-up $10
19. $35, mark-up $15
20. $80, mark-up $20
21. $150, mark-up $50
22. $12, mark-up $8
23. $250, mark-up $100
24. $64, mark-up $16

**ROUND 4** Questions 25–32: take the discount off the price to find the sale price.

25. $50, discount $10
26. $80, discount $20
27. $45, discount $5
28. $120, discount $30
29. $200, discount $40
30. $99, discount $9
31. $300, discount $75
32. $36, discount $12

## Level 2 · Building up

Medium · 28 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Use a calculator. Show the sum you type in.**

**a** Cost $40, mark-up 25%. Find the selling price.

**b** Price $60, discount 20%. Find the sale price.

**c** Cost $50, sold $60. Find the profit.

**d** Write the profit in c as a % of the cost price.

**YOU DO** On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

28 QUESTIONS

**REMEMBER** Mark-up: find the % of the cost price, then add it on · Discount: find the % of the price, then take it off · % profit = profit ÷ cost price × 100.

### ROUND 1 Questions 1–8: find the selling price after each mark-up.

1. $48, mark-up 25%
2. $80, mark-up 10%
3. $60, mark-up 50%
4. $200, mark-up 20%
5. $120, mark-up 5%
6. $36, mark-up 25%
7. $90, mark-up 30%
8. $64, mark-up 15%

### ROUND 2 Questions 9–16: find the sale price after each discount.

9. $70, discount 20%
10. $80, discount 25%
11. $150, discount 10%
12. $45, discount 20%
13. $300, discount 40%
14. $96, discount 25%
15. $250, discount 30%
16. $88, discount 15%

### ROUND 3 Questions 17–24: find the profit as a percentage of the cost price.

17. Cost $20, sold $25
18. Cost $50, sold $80
19. Cost $40, sold $44
20. Cost $80, sold $120
21. Cost $200, sold $230
22. Cost $10, sold $13
23. Cost $25, sold $35
24. Cost $30, sold $60

### ROUND 4 Questions 25–28: solve each problem. Show your working.

25. A shop pays $7.60 for a pen. It marks the pen up by 50%. Find the selling price and the profit.

26. An airfare costs $7 000. It is discounted by 40%. What is the sale price?

27. A shirt was $56. It is discounted by 26%. What is the sale price?

28. Kim buys a bike for $80 and sells it for $116. Find the profit as a percentage of the cost price.

LESSON 1.4

# Wages and overtime

Date _____

**Learning intention**
We are learning to work out pay from an hourly rate or a yearly salary, and add on overtime pay.

**Success criteria**
- ☐ I can find the pay for some hours of work
- ☐ I can change pay between weekly, fortnightly and yearly
- ☐ I can work out overtime pay, and the total pay

## Level 1 · Getting started

Easy · 32 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** **Pay = hourly rate × hours. 1 fortnight = 2 weeks. 1 year = 52 weeks.**

**a** $20 an hour for 8 hours. Find the pay.

**b** $600 a week. Find the pay for a fortnight.

**c** $600 a week. Find the pay for a year.

**d** $52 000 a year. Find the pay for a week.

**YOU DO** On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

32 QUESTIONS

**REMEMBER** Pay = hourly rate × hours · Week to fortnight: × 2 · Week to year: × 52 · Year to week: ÷ 52.

**ROUND 1** Questions 1–8: find the pay (hourly rate × hours).

1. $20 an hour for 5 hours
2. $25 an hour for 6 hours
3. $18 an hour for 10 hours
4. $30 an hour for 7 hours
5. $20 an hour for 38 hours
6. $15 an hour for 8 hours
7. $25 an hour for 40 hours
8. $24 an hour for 10 hours

**ROUND 2** Questions 9–16: find the pay for a fortnight (weekly pay × 2).

9. $400 a week
10. $550 a week
11. $600 a week
12. $750 a week
13. $900 a week
14. $1 000 a week
15. $820 a week
16. $1 250 a week

**ROUND 3** Questions 17–24: find the pay for a year (weekly pay × 52).

17. $100 a week
18. $500 a week
19. $1 000 a week
20. $200 a week
21. $600 a week
22. $800 a week
23. $300 a week
24. $750 a week

**ROUND 4** Questions 25–32: find the pay for a week (yearly salary ÷ 52).

25. $52 000 a year
26. $26 000 a year
27. $104 000 a year
28. $78 000 a year
29. $65 000 a year
30. $39 000 a year
31. $57 200 a year
32. $91 000 a year

## Level 2 · Building up

Medium · 24 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Ari earns $20 an hour. Find Ari's pay for each part.**

**a** 1 hour at time and a half (× 1.5)

**b** 1 hour at double time (× 2)

**c** 4 hours at time and a half

**d** 38 hours at $20 + 4 hours at time and a half

**YOU DO** On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 24 QUESTIONS

**REMEMBER** Time and a half = 1.5 × rate · Double time = 2 × rate · Total pay = normal pay + overtime pay.

**ROUND 1** Questions 1–8: find the pay for 1 hour of overtime. Rates are per hour.

1. $20, time and a half
2. $20, double time
3. $30, time and a half
4. $17, double time
5. $24, time and a half
6. $25, double time
7. $16, time and a half
8. $25, time and a half

**ROUND 2** Questions 9–16: find the pay for 4 hours of overtime. Rates are per hour.

9. $20, time and a half
10. $30, double time
11. $16, time and a half
12. $25, double time
13. $24, time and a half
14. $35, double time
15. $22, time and a half
16. $40, double time

**ROUND 3** Questions 17–20: find the total pay (normal pay + overtime pay). Rates are per hour.

17. 38 hours at $25 + 2 hours at time and a half
18. 35 hours at $30 + 3 hours at double time
19. 40 hours at $20 + 6 hours at time and a half
20. 30 hours at $18 + 5 hours at double time

**ROUND 4** Questions 21–24: solve each problem. Show your working.

21. Lee works 30 hours at $15.78 an hour. What is Lee's gross income (total pay)?
22. Jo earns $30 an hour. Jo works 6 hours at double time. Find Jo's pay.
23. Mia earns $24 an hour. One week, Mia works 38 normal hours and 5 hours at time and a half. Find Mia's total pay.
24. Bill works 3 day shifts and 5 night shifts of 8 hours. Day shifts pay $21.20 an hour. Night shifts pay $33.56 an hour. Find Bill's total pay.

LESSON 1.5

# Commission, gross and net income

Date _____

**Learning intention**
We are learning to work out commission, gross income and net income.

**Success criteria**
- ☐ I can find the commission on some sales
- ☐ I can find the gross income (all the money earned)
- ☐ I can take away the deductions to find the net income

## Level 1 · Getting started

Easy · 28 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** **Commission = % of sales. Retainer = set weekly pay. Gross income = total earned.**

**a** Find 10% commission on sales of $800.

**b** Find 5% commission on sales of $2 000.

**c** Gross income: $300 retainer + $150 commission.

**d** Gross income: $250 wage, $40 overtime, $75 commission.

**YOU DO** On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

28 QUESTIONS

**REMEMBER** Commission: 10% of $800 = 0.1 × 800 = $80 (5% → 0.05) · Gross income: add up all the money earned.

**ROUND 1** Questions 1–8: find the 10% commission on each amount of sales.

1. Sales of $700
2. Sales of $500
3. Sales of $2 000
4. Sales of $300
5. Sales of $1 500
6. Sales of $4 000
7. Sales of $650
8. Sales of $12 000

**ROUND 2** Questions 9–16: find the commission on each amount of sales.

9. 5% of $2 000 sales
10. 20% of $450 sales
11. 5% of $600 sales
12. 25% of $800 sales
13. 10% of $950 sales
14. 2% of $1 000 sales
15. 50% of $300 sales
16. 5% of $10 000 sales

**ROUND 3** Questions 17–24: find the gross income (add the two amounts).

17. Wage $500 Commission $80
18. Retainer $300 Commission $150
19. Wage $600 Commission $150
20. Retainer $250 Commission $400
21. Wage $750 Commission $90
22. Retainer $200 Commission $85
23. Wage $820 Commission $75
24. Retainer $450 Commission $125

**ROUND 4** Questions 25–28: find the gross income (add up all the money earned).

25. Wage $500, overtime $60, commission $40
26. Wage $400, overtime $100, commission $200
27. Wage $600, overtime $90, commission $50
28. Wage $350, overtime $75, commission $125

## Level 2 · Building up

Medium · 24 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Use a calculator. Net income = gross income – deductions (tax, super, union fees).

**a** Find 3.5% commission on sales of $4 600.

**b** Gross income: $400 retainer + 5% of $3 000 sales.

**c** Net income: $900 gross, $180 tax, $90 super.

**d** Gross and net pay: 38 hours at $25 an hour, $190 tax.

**YOU DO** On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 24 QUESTIONS

**REMEMBER** Commission = % as a decimal × sales (4.5% → 0.045) · Net income = gross income – deductions · Deductions: tax, super (superannuation), union fees.

### ROUND 1 Questions 1–8: use a calculator to find the commission.

1. 3% of $1 250 sales
2. 2.5% of $8 000 sales
3. 1.5% of $24 000 sales
4. 6% of $735 sales
5. 12% of $480 sales
6. 3.5% of $2 600 sales
7. 7.5% of $940 sales
8. 4% of $1 895 sales

### ROUND 2 Questions 9–16: find the gross income (retainer + commission).

9. Retainer $300 plus 10% of $2 000
10. Retainer $250 plus 5% of $4 000
11. Retainer $400 plus 2% of $6 500
12. Retainer $500 plus 3% of $1 800
13. Retainer $350 plus 4% of $2 250
14. Retainer $200 plus 8% of $1 500
15. Retainer $450 plus 1.5% of $12 000
16. Retainer $320 plus 2.5% of $3 600

### ROUND 3 Questions 17–20: find the net income (gross income – deductions).

17. Gross $1 200. Tax $250, super $110.
18. Gross $750. Tax $120, super $70.
19. Gross $1 000. Tax $215.50, super $95.
20. Gross $820. Tax $96.40, super $78.

### ROUND 4 Questions 21–24: solve each problem. Show your working.

21. Pedro gets 4.5% commission on sales of $790. How much commission does Pedro get?
22. Alex gets a $500 retainer plus 2% commission. Sales are $45 000. Find Alex's gross income.
23. Jess earns $1 200. Jess pays $230 tax, $114 super and $15 union fees. Find Jess's net income.
24. Tranh works 36 hours at $28.89 an hour. Tax $142.59, super $22.50. Find gross and net pay.

LESSON 1.6

# Simple interest and loans

Date _____

**Learning intention**
We are learning to work out simple interest, and the cost of paying off a loan.

**Success criteria**
- ☐ I can write an interest rate as a decimal
- ☐ I can use I = PRN to find the simple interest
- ☐ I can find the deposit, total repaid and interest on a loan

## Level 1 · Getting started

Easy · 28 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** **I = P × R × N. P = money invested, R = rate as a decimal, N = number of years.**

**a** Write 5% as a decimal.

**b** Find the simple interest on $1 000 at 5% for 2 years.

**c** Find the simple interest on $400 at 10% for 3 years.

**d** $500 is invested at 10% for 2 years. Find the total at the end.

**YOU DO** On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

28 QUESTIONS

**REMEMBER** Rate as a decimal: ÷ 100 (5% → 0.05) · Simple interest: I = P × R × N · Total at the end = P + I.

**ROUND 1** Questions 1–4: write each interest rate as a decimal.

1. 5%
2. 10%
3. 3%
4. 20%

**ROUND 2** Questions 5–12: find the simple interest for 1 year.

5. $200 at 10%
6. $500 at 10%
7. $100 at 5%
8. $600 at 5%
9. $1 000 at 2%
10. $300 at 20%
11. $800 at 10%
12. $2 000 at 4%

**ROUND 3** Questions 13–20: find the simple interest.

13. $1 000 at 5% for 2 years
14. $500 at 10% for 3 years
15. $200 at 10% for 4 years
16. $400 at 5% for 2 years
17. $2 000 at 3% for 5 years
18. $700 at 10% for 2 years
19. $1 000 at 4% for 3 years
20. $300 at 20% for 3 years

**ROUND 4** Questions 21–28: find the total at the end (money invested + interest).

21. $500 at 10% for 2 years
22. $1 000 at 5% for 3 years
23. $200 at 10% for 5 years
24. $400 at 5% for 4 years
25. $2 000 at 2% for 3 years
26. $800 at 10% for 3 years
27. $100 at 20% for 2 years
28. $3 000 at 5% for 2 years

## Level 2 · Building up

Medium · 20 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Use a calculator. Monthly repayments: number of repayments = 12 × years.**

**a** Find the simple interest on $2 500 at 4.5% for 3 years.

**b** A $3 000 loan is repaid at $150 a month for 2 years. Find the total repaid.

**c** Find the interest paid on the loan in part b.

**d** A $2 000 bike: 10% deposit, then $90 a month for 2 years. Find the interest.

**YOU DO** On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

20 QUESTIONS

**REMEMBER** I = P × R × N (4.5% → 0.045) · Total repaid = deposit + all the repayments · Interest = total repaid − price.

**ROUND 1** Questions 1–4: use a calculator to find the simple interest (6 months = 0.5 years).

1. $1 250 at 3.5% for 3 years
2. $5 000 at 8% for 4 years
3. $645 at 5% for 2 years
4. $800 at 5% for 6 months

**ROUND 2** Questions 5–8: find the deposit (a % of the price).

5. Price $2 500, 10% deposit
6. Price $1 800, 20% deposit
7. Price $4 000, 15% deposit
8. Price $3 600, 25% deposit

**ROUND 3** Questions 9–12: find the total repaid (repayment × 12 × years).

9. $200 a month for 5 years
10. $250 a month for 3 years
11. $110 a month for 3 years
12. $95.50 a month for 2 years

**ROUND 4** Questions 13–16: find the interest paid (total repaid − loan).

13. Loan $2 000. Repay $100 a month for 2 years.
14. Loan $1 000. Repay $50 a month for 2 years.
15. Loan $4 000. Repay $150 a month for 3 years.
16. Loan $3 500. Repay $120 a month for 3 years.

**ROUND 5** Questions 17–20: solve each problem. Show your working.

17. Kim invests $600 at 5% simple interest for 4 years. How much interest does Kim earn?
18. Jo borrows $5 000. Jo repays $200 a month for 5 years. How much interest does Jo pay?
19. Ana buys a $1 200 phone. Ana pays a 10% deposit, then $50 a month for 2 years. Find the deposit and the total paid.
20. A car is worth $7 000. Lee pays a 15% deposit, then $250 a month for 4 years. Find the deposit, the total repaid and the interest.

LESSON 1.7

# Compound interest

Date _____

**Learning intention**
We are learning to work out compound interest, year by year and with the formula $A = P(1 + R)^n$.

**Success criteria**
- ☐ I can find the interest for a year and the new balance
- ☐ I can compare compound interest with simple interest
- ☐ I can find the balance and interest with $A = P(1 + R)^n$

## Level 1 · Getting started

Easy · 22 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** **$1 000 earns 10% compound interest. Each year, find the interest, then add it on.**

**a** Find the interest for year 1.

**b** Find the balance after 1 year.

**c** Find the balance after 2 years.

**d** Find the balance after 2 years with simple interest.

**YOU DO** On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

22 QUESTIONS

**REMEMBER** Interest = rate × balance (10% → × 0.1) · New balance = balance + interest · Repeat each year.

**ROUND 1** Questions 1–4: find the interest for 1 year.

1. $100 at 10%
2. $1 000 at 20%
3. $400 at 25%
4. $2 000 at 5%

**ROUND 2** Questions 5–12: find the balance after 1 year (balance + interest).

5. $300 at 10%
6. $1 000 at 10%
7. $400 at 5%
8. $500 at 20%
9. $200 at 25%
10. $700 at 10%
11. $3 000 at 5%
12. $600 at 20%

**ROUND 3** Questions 13–20: find the balance after 2 years at 10% compound interest.

13. $100
14. $200
15. $500
16. $1 000
17. $300
18. $2 000
19. $400
20. $600

**ROUND 4** Questions 21–22: fill in each table. Then answer the question on the line.

21. $2 000 earns 10% compound interest for 3 years. Find the total interest.

| Year | 1 | 2 | 3 |
|---|---|---|---|
| Interest | | | |
| Balance | | | |

22. $500 earns 20% a year for 3 years. How much more does compound interest give?

| Balance after year | 1 | 2 | 3 |
|---|---|---|---|
| Simple interest | | | |
| Compound interest | | | |

## Level 2 · Building up

Medium · 28 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Use a calculator. $A = P(1 + R)^n$, R = rate as a decimal, n = number of years.

**a** Write (1 + R) for a rate of 6%.

**b** Find A for $1 000 at 10% for 3 years.

**c** Find A for $2 500 at 6% for 4 years, to the nearest cent.

**d** Find the interest earned in part c.

**YOU DO** On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

28 QUESTIONS

**REMEMBER** $A = P \times (1 + R)^n$ · 6% → 1.06 · Type 2500 × 1.06 ^ 4, then press = · Round to the nearest cent · Interest = A − P.

### ROUND 1 Questions 1–8: write (1 + R) as a decimal for each rate (5% → 1.05).

1. 10%
2. 3%
3. 20%
4. 8%
5. 12%
6. 4%
7. 2.5%
8. 4.5%

### ROUND 2 Questions 9–16: use $A = P(1 + R)^n$ to find the balance. Round to the nearest cent.

9. $1 000 at 10% for 2 years
10. $500 at 10% for 3 years
11. $2 000 at 5% for 3 years
12. $800 at 20% for 2 years
13. $1 500 at 4% for 2 years
14. $600 at 3% for 4 years
15. $4 000 at 2.5% for 3 years
16. $2 500 at 6% for 5 years

### ROUND 3 Questions 17–24: find the compound interest earned (A − P).

17. $1 000 at 5% for 2 years
18. $3 000 at 10% for 2 years
19. $400 at 20% for 3 years
20. $2 000 at 3% for 3 years
21. $5 000 at 8% for 2 years
22. $1 200 at 4% for 3 years
23. $750 at 6% for 4 years
24. $10 000 at 2% for 5 years

### ROUND 4 Questions 25–28: solve each problem. Show your working.

25. Ella invests $5 000 at 4% compound interest for 3 years. Find the value and the interest earned.
26. Kai invests $4 600 at 12% compound interest for 2 years. How much interest does Kai earn?
27. Jess invests $2 000 at 5% for 3 years. Find the simple interest and the compound interest.
28. Max borrows $5 000 at 9.75% compound interest for 2 years. How much interest does Max pay?